

Answer any two questions from the following questions

1. What are Divide and Conquer algorithms? Describe how they work. Can you give any common examples of the types of problems where this approach might be used?
2. What are the Dijkstra and Prim algorithms, and how are they implemented?
3. A numeric array of length N is given. We need to design a function that finds all positive numbers in the array that have their opposites in it as well. Describe approaches for solving optimal worst case and optimal average case performance, respectively.

Course Code: CCE- 411

Time: 50 minutes

Course Title: Algorithm Engineering

Marks: 15

1. Define NP-complete problem. What is reduction? How to solve NP-complete problems? 4
2. Write the formal definition of the approximation algorithm. Design an approximation algorithm of vertex cover. 4
3. Use the following example to Calculate Time and Space Complexity from an Algorithm in Frequency Count Method? 4

```
AlgorithmMultiply (A,B,n)
{
  for(int i=0;i<n;i++)
  {
    for (int j=0;j<n;j++)
    {
      C[i,j]=0;
      for(int k=0;k<n;k++)
      {
        C[i,j]=C[i,j]+ A[i,k]* B[k,j];
      }
    }
  }
}
```

4. How to make heuristic function from the following start state to goal state? 3

7	2	4
5	-	6
8	3	1

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Final Examination of B. Sc. Engineering in CSE Level: 4 Semester: I Session: 2017-2018

Course Code
CCE-411

Course Title
Algorithm Engineering

January-June 2021

Credit: 03
Time: 03 Hr
Marks: 70

Answer any 05 out of 06 Questions (Split answers are highly discouraged)

[A.] i. What is meant by computational complexity? Which is the best algorithm for optimization? 7

ii. Why optimization algorithm is used? Where can optimization be used?
[B.] How to Calculate Time and Space Complexity from an Algorithm in Frequency Count Method? Use the following example to calculate. 7

Algorithm Multiply (A,B,n)

```
{
for(int i=0; i<n; i++) n+1
```

```
{
for (int j=0; j<n; j++) n(n+1)
```

```
{
C[i,j]=0; n*n
```

```
for(int k=0; k<n; k++) n*n*n(n+1)
```

```
{
C[i,j]=C[i,j]+ A[i,k]* B[k,j]; n*n*n*n
}
```

```
}
}
```

$$= n+1$$

$$= n^2 + n$$

$$= n^3 + n^2 + n$$

$$= \frac{n^3 + 3n^2 + 2n + 1}{2}$$

$3n^3 + 4$

[A.] How we can find Optimization: Minima and Maxima? Explain with proper example. 7

[B.] What is Heuristic Search? What is the difference between algorithm and a heuristic? What is an example of a heuristic? 7

[A.] How to make heuristic function from the following start state to goal state in ascending order? 7

7	2	4
5	-	6
8	3	1

-	1	2
3	9	5
6	7	8

[B.]

a. What are the types of linear programming? 7

b. Calculate the maximal and minimal value of $z = 5x + 3y$ for the following constraints using Graphical Method.

$$x + 2y \leq 14$$

$$3x - y \geq 0$$

$$x - y \leq 2$$

$$x + \frac{1}{2}y = 7$$

$$= \frac{3x}{2} = 9$$

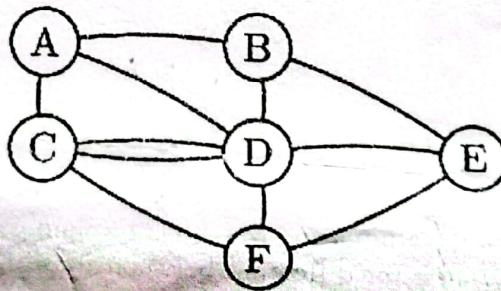
$$5x = \frac{18}{2} = 9$$

$$x - 3x = -2$$

$$\therefore x = 2$$

Column answers

- 4 [A.] What is complexity class? Compare NP-complete and NP-hard complexity classes with related examples. 3
- [B.] Define optimization problems. How solve NP-complete problems including brute-force algorithm, heuristics, and approximation algorithms? 3
- [C.] Write the formal definition of the approximation algorithm. Explain the following statement: "approximation algorithms are typically used when finding an optimal solution is intractable". 4
- [D.] What is approximation ratio? Suppose you are familiar with vertex cover approximation algorithm and maximal matching. Prove the following statement: $|C| = 2 * |M| \leq 2 * |C^*|$, where C is a vertex cover, M is a maximal matching, and C^* is an optimal cost. 4
- 5 [A.] Define fixed parameter tractability. Design a fixed parameter- k bounded search tree algorithm. How to analyze the complexity of bounded search tree algorithm? 6
- [B.] What is the function of kernelization for designing an efficient algorithm? How to kernelize the giant problem? 4
- [C.] What is a heuristic technique? Mention the limitation of the heuristic technique for obtaining a solution. 4
- 6 [A.] What are the advantages of randomized algorithm beyond deterministic algorithm? 3
- [B.] Distinguish between the Las Vegas algorithms and Monte Carlo algorithms with examples. Show the complexity of randomized Quick sort algorithm is $O(n \log n)$ using related example and parameters. 6
- [C.] What is minimum cut problem? Describe the solution of minimum cut problem in the given graph using the contract algorithm. 5



Correct
Answer

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Final Examination of B. Sc. Engineering in CSE Level: IV Semester: I Session: 2016-2017

Course Code
CCE-411

Course Title
Algorithm Engineering

January-June 2020

Credit: 03
Time: 03 Hr
Marks: 70

Answer any 05 out of 06 Questions (Split answers are highly discouraged)

- 1) [A.] Give a real-world example that requires sorting. — 3
 [B.] Write pseudocode for insertion sort with an example 8
 [C.] Come up with a real-world problem in which only the best solution will do. Then come up with one in which a solution that is "approximately" the best is good enough. 3

- 2 [A.] Explain a solution of the maximum-subarray problem using the divide-and-conquer technique. 6
 [B.] Write pseudocode for Strassen's algorithm with an example. 6
 [C.] Write short notes on divide, conquer, combine and recurrences. 2

- 3 [A.] Explain worst case analysis and probabilistic analysis of Hiring Problem. 5
 [B.] What is the running time of HEAPSORT on an array A of length n that is already sorted in increasing order? What about decreasing order? 5
 [C.] How would you modify QUICKSORT to sort into non increasing order? 4

- 4 [A.] Write pseudocode for BFS algorithm. Write also the time complexity of the BFS. $O(V+E)$ 7
 [B.] What is the running time of BFS if we represent its input graph by an adjacency matrix and modify the algorithm to handle this form of input? 3
 [C.] Rewrite the procedure DFS, using a stack to eliminate recursion. — 4
 (2)

- 5 [A.] Write pseudocode for Prim algorithm. 7
 [B.] Kruskal's algorithm can return different spanning trees for the same input graph G, depending on how it breaks ties when the edges are sorted into order. Show that for each minimum spanning tree T of G, there is a way to sort the edges of G in Kruskal's algorithm so that the algorithm returns T. 10
 [C.] Write down the difference between NP hard and NP complete problem. 3
 (4)

- 6 [A.] Write pseudocode for Bellman-Ford algorithm. 7
 [B.] Give a simple example of a directed graph with negative-weight edges for which Dijkstra's algorithm produces incorrect answers. 4
 [C.] Proof that, satisfiability of Boolean formulas in 3-conjunctive normal form is NP-complete 3
 (5)

Patuakhali Science and Technology University

7th semester (Level-4, Semester-I) Final Examination of B.Sc. Engg. (CSE), January-June 2019, Session 2015-2016

Course Code: CCE-411

Course Title: Algorithm Engineering

Credit Hours: 3.0

Full Marks: 70 Duration: 3 Hours

[Figures in the right margin indicate full marks. **Split answering of any question is prohibited**]

Answer any 5 of the following questions.

1. (a) Why analysis of algorithm is important to develop an efficient solution? 2
 (b) Illustrate the decision problems as the central objects of study in computational complexity. 3
 (c) Mention the differences between deterministic and non-deterministic algorithms. 3
 (d) Is polynomial time desired to formulate a solution of a problem? 2
 (e) What are P, NP, NP-Complete, and NP-Hard problems? 4
2. (a) Define parameterized complexity. How to formulate a fixed-parameterized tractability problem? 2+2=4
 (b) Describe an efficient algorithm for BAR FIGHT PREVENTION problem with $k=3$. 4
 (c) Why kernelization is important tool to design parameterized algorithm? 2
 (d) Illustrate the kernelization technique by obtaining kernel for vertex cover. 4
3. (a) Define dominating set of a graph. State the steps of an efficient approximation algorithm for finding the smallest dominating set for all graphs. 2+3=5
 (b) Mention some NP-hard problems that are FPT. 2
 (c) What are the partitioning parts of crown decomposition? Illustrate an example for obtaining a kernel via crown decomposition. 2+2=4
 (d) State the steps of branching algorithm for building bounded search tree. 3
4. (a) What is combinatorial optimization? 2
 (b) Formulate the problem of Minimum Spanning Tree (MST). (Give some real applications of MST). 2+1=3
 (c) Explain the Kruskal's algorithm for finding the MST. What is the time complexity of this algorithm? 4+2=6
 (d) Briefly explain the three properties of maximum flow problem. 3
5. (a) What are the different variants of shortest path problem? How to formulate single source shortest path (SSSP) problem? 2+2=4
 (b) How to compute edge relaxation for obtaining shortest path? 3
 (c) State the Bellman-Ford algorithm for the SSSP problem and explain this algorithm by using arbitrary cost functions of a given graph. How to improve the running time of this algorithm? 5+2=7
6. (a) How to cope with the hardness results of combinatorial optimization problems by using approximation algorithms? 2
 (b) Find an optimal solution by using an approximation scheme for Knapsack problem where $n=7$, $m=15$, $p(10, 5, 15, 7, 6, 18, 3)$, and $w(2, 3, 5, 7, 1, 4, 1)$ respectively. 4
 (c) What is randomized algorithm? How to analyze randomized algorithm? 2+2=4
 (d) Discuss the randomized version of QuickSort and analyze the time complexity of this algorithm. 4

CSE faculty, PSTU. Mid exam of CCE-411: Algorithm Engineering (7th semester, Session: 2015-16)
Time: 50 minutes, Marks: 15

1. What is computational complexity? Explain the notion of computational efficiency with examples. (2+2=4)
2. When Asymptotic Notations are used to analyze the algorithms? Distinguish $O(N^2)$ and $O(2^N)$ with examples. (1+2=3)
3. What are NP complete and NP hard problems? (2=2)
4. When a parameterized problem is called fixed parameter tractable? Illustrate the technique by obtaining kernel for vertex cover. (1+3=4)
5. Briefly explain the crown decomposition with an example. (2=2)